

HIV — Transmission, U=U & Prevention

LEFT SIDE

EITHER ONE ALONE = **AIDS**

AIDS STAGING VERDICT

AIDS DIAGNOSIS
— CDC STAGE 3

LEFT GATE
<200
cells/ μ L

— GATE 1 —
CD4 <200
— REGARDLESS OF SYMPTOMS
— IN ANYONE >6 YEARS OLD

AIDS STAGING VERDICT

— GATE 2 —
ANY AIDS-DEFINING
ILLNESS
— REGARDLESS OF
CD4 COUNT

CENTER WALL

AIDS-DEFINING ✓

- PCP
- TOXOPLASMA
- CRYPTOCOCCUS
- CMV RETINITIS
- KAPOSI'S SARCOMA
PURPLE LESION
- NON-HODGKIN'S
LYMPHOMA
- CERVIX WITH INVASIVE
CARCINOMA
- MAC BACTERIUM
- PML
— PROGRESSIVE
MULTIFOCAL
LEUKOENCEPHALOPATHY
- ESOPHAGEAL
CANDIDIASIS
- WASTING PATIENT
- HIV BRAIN
ENCEPHALOPATHY
- TB

AIDS-DEFINING ✓

NOT AIDS-DEFINING ✗

- HODGKIN'S
LYMPHOMA
= NOT AIDS-DEFINING
= INCREASED RISK
BUT NOT DEFINING
- MELANOMA
- LUNG/LIVER/
ANAL CANCER
- MULTIPLE
MYELOMA
- LEUKEMIA
- ZOSTER IN SINGLE
DERMATOME
= NOT AIDS-DEFINING
= BUT SIGNALS
EARLY IMMUNE
DECLINE

NOT AIDS-DEFINING ✗

★ KS =
NOT STRICTLY
CD4-DEPENDENT
— CAN APPEAR
WITH
NORMAL CD4

RIGHT SIDE

TRANSMISSION RISK LEAGUE TABLE

ROUTE	TOP SPOT	ROUTES
BLOOD TRANSFUSION	9,250/10,000 = 92.5%	HIGHEST RISK
RECEPTIVE ANAL INTERCOURSE	138/10,000 — ~1.4%	THIN RECTAL MUCOSA — MICROTRAUMA — LANGERHANS CELLS DIRECTLY BELOW SURFACE
NEEDLE SHARING	63/10,000 — ~0.6%	
NEEDLE STICK HEALTH WORKER	23/10,000 — ~0.23%	
RECEPTIVE VAGINAL INTERCOURSE	8/10,000 —	MALE TO FEMALE MORE EFFICIENT
INSERTIVE ANAL INTERCOURSE	11/10,000	
INSERTIVE VAGINAL INTERCOURSE	4/10,000 —	FEMALE TO MALE LESS EFFICIENT
BITING/SPITTING/TEARS	NEGLECTIBLE	— NOT SIGNIFICANT

U=U

UNDETECTABLE =
UNTRANSMITTABLE

VIRAL LOAD
<20 COPIES/mL
→
ESSENTIALLY
ZERO SEXUAL
TRANSMISSION

>125,000 SEX
ACTS WITHOUT
TRANSMISSION

MTCT PREVENTION STATION

— ART THROUGHOUT
PREGNANCY AND
DELIVERY

★ MOST EFFECTIVE
— SUPPRESSES VIRAL LOAD

— ANTIRETROVIRAL
PROPHYLAXIS FOR
NEWBORN 4–6 WEEKS

— CESAREAN SECTION
REDUCES BIRTH CANAL EXPOSURE

— AVOID BREASTFEEDING

WITHOUT
INTERVENTION

INTERVENTION =
15–35%
TRANSMISSION
WITH ALL
INTERVENTIONS
= <1%

RIGHT SIDE

TRANSMISSION RISK LEAGUE TABLE	
ROUTE	ROUTES
TOP SPOT	 BLOOD TRANSFUSION 9,250/10,000 = 92.5% — HIGHEST RISK 
♀	RECEPTIVE ANAL INTERCOURSE 138/10,000 — ~1.4% — THIN RECTAL MUCOSA — MICROTRAUMA — LANGERHANS CELLS DIRECTLY BELOW SURFACE

Transmission Risk League Table

WHAT YOU SEE

A wooden 'TRANSMISSION RISK LEAGUE TABLE' ranking exposure routes top-to-bottom, the top row in green marked TOP SPOT — the league-table layout literally ranks per-exposure HIV risk from highest to lowest.

WHAT IT ENCODES

Per-exposure HIV transmission risk varies by orders of magnitude by route, so board questions often ask you to rank exposures. Descending order to know: blood transfusion (~90%+) >> perinatal/needle-sharing > receptive anal > needlestick > receptive vaginal > insertive routes >> bites/saliva (negligible). Risk is amplified by high source viral load, genital ulcers/STIs, and breaches in mucosa.

THE BOARD TRAP

WRONG: 'Casual contact (hugging, sharing utensils, toilet seats, mosquitoes) can transmit HIV.' These do NOT transmit HIV — the virus requires direct exposure of blood/genital secretions/breast milk to mucosa or bloodstream. The discriminator: only blood, sexual fluids, and breast milk are infectious routes; environmental/casual contact is not.

TRANSMISSION RISK LEAGUE TABLE	
ROUTE	ROUTES
TOP SPOT	 BLOOD TRANSFUSION 9,250/10,000 = 92.5% — HIGHEST RISK 
♀	RECEPTIVE ANAL INTERCOURSE 138/10,000 — ~1.4% — THIN RECTAL MUCOSA — MICROTRAUMA — LANGERHANS CELLS DIRECTLY BELOW SURFACE
	NEEDLE SHARING 63/10,000 — ~0.6%
	NEEDLE STICK HEALTH WORKER

Blood transfusion — top risk

WHAT YOU SEE

The green TOP-SPOT row of the league table: a blood-bag icon labelled BLOOD TRANSFUSION, $9,250/10,000 = 92.5\%$ HIGHEST RISK — sitting at #1 because a whole infected unit delivered IV is near-certain transmission.

WHAT IT ENCODES

Transfusion of HIV-infected blood is the single highest per-exposure transmission risk, roughly 90%+ (about $9,250+/10,000$ acts). The huge inoculum delivered directly intravascularly accounts for the near-certainty of transmission. This is why blood-bank screening (antibody, p24 antigen, and nucleic acid testing) is universal and has made transfusion-related HIV extremely rare in resource-rich settings.

THE BOARD TRAP

WRONG: 'Transfusion is no longer a meaningful HIV risk so it ranks low.' Although rare today due to NAT/antigen donor screening, transfusion remains the HIGHEST per-act transmission risk when an infected unit is given. Ranking it low confuses real-world frequency (very low) with per-exposure efficiency (highest) — the question is about per-act risk.

N'S DMA DEFINING RISK FINING A R/ CER	TOP SPOT	 BLOOD TRANSFUSION 9,250/10,000 = 92.5% — HIGHEST RISK 	 UNDETECTABLE UNTRANSMITTABLE — VIRAL LOAD <20 COPIES/mL —> ESSENTIAL ZERO SEXUAL TRANSMISSION — >125,000 S ACTS WITH
		RECEPTIVE ANAL INTERCOURSE 138/10,000 — ~1.4% — THIN RECTAL MUCOSA — MICROTRAUMA — LANGERHANS CELLS DIRECTLY BELOW SURFACE	
		NEEDLE SHARING 63/10,000 — ~0.6%	
		NEEDLE STICK HEALTH WORKER 23/10,000 — ~0.23%	
		RECEPTIVE VAGINAL INTERCOURSE 8/10,000 — MALE TO FEMALE MORE EFFICIENT	

Receptive anal intercourse

WHAT YOU SEE

League-table row RECEPTIVE ANAL INTERCOURSE
 138/10,000 ~1.4% with a note 'thin rectal mucosa, microtrauma, Langerhans cells below surface' — the highest sexual-route rung because that fragile mucosa gives the virus direct access.

WHAT IT ENCODES

Receptive anal intercourse is the highest-risk SEXUAL exposure (~138/10,000 acts, ~1.4%) because the rectal mucosa is thin, single-layered, richly vascular, and easily traumatized, giving the virus direct access to target cells. Risk rises further with ejaculation, mucosal tears, concurrent STIs, and high source viral load.

THE BOARD TRAP

WRONG: 'Insertive and receptive partners share equal risk.'
 The receptive partner is at far higher risk than the insertive partner for any sexual act (anal or vaginal). The discriminator is mucosal exposure: receptive >> insertive. Treating risk as symmetric between partners is the trap.

A R/ CER			+	— LANGERHANS CELLS DIRECTLY BELOW SURFACE	— VIRAL LOAD <20 COPIES/ → ESSENTIAL ZERO SEXU TRANSMISSI — >125,000 S ACTS WITHO TRANSMISSI
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				INSERTIVE VAGINAL INTERCOURSE 4/10,000 — FEMALE TO MALE LESS EFFICIENT	
				BITING/SPITTING/TEARS NEGLECTIBLE — NOT SIGNIFICANT	

Needlestick — occupational

WHAT YOU SEE

League-table rows for a needle: NEEDLE SHARING 63/10,000 ~0.6% and NEEDLE STICK HEALTH WORKER 23/10,000 ~0.23% — the occupational hollow-needle exposure that triggers PEP.

WHAT IT ENCODES

Occupational percutaneous needlestick from an HIV-positive source carries ~0.3% risk (~23/10,000). This is the trigger for occupational post-exposure prophylaxis (PEP): start a 3-drug ART regimen ASAP, ideally within hours and no later than 72 hours, and continue for 28 days. Baseline and follow-up HIV testing of the exposed worker is required.

THE BOARD TRAP

WRONG: 'Needlestick risk is the same regardless of injury type.' Risk is modified by depth of injury, a hollow-bore (vs solid suture) needle, visible blood on the device, and high source viral load. A deep stick with a blood-filled hollow needle from a high-VL source is much higher risk than a superficial solid-needle injury — these factors guide PEP urgency.



Receptive vaginal intercourse

WHAT YOU SEE

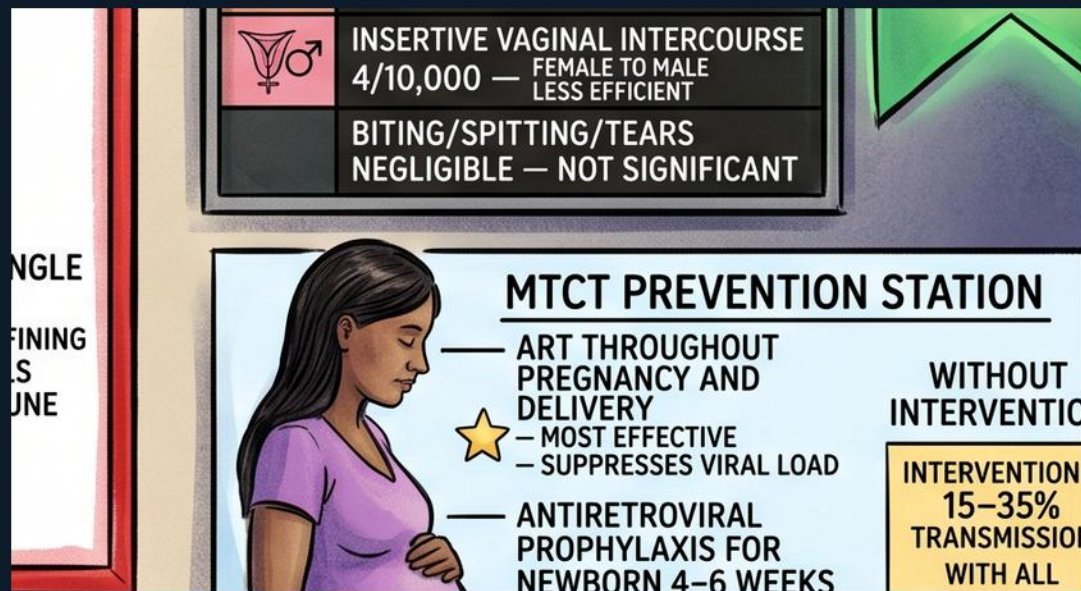
League-table row RECEPTIVE VAGINAL INTERCOURSE (male-to-female) 8/10,000 'more efficient', sitting above INSERTIVE VAGINAL 4/10,000 — receptive partner is twice the risk of the insertive partner.

WHAT IT ENCODES

Receptive vaginal intercourse carries higher risk (~8/10,000) than insertive vaginal intercourse (~4/10,000). The vaginal/cervical mucosa is more exposed than the insertive partner's, and microabrasions increase transmission. Risk is heavily modulated by source viral load (U=U eliminates sexual risk when undetectable).

THE BOARD TRAP

WRONG: 'Vaginal transmission is unaffected by other infections.' Co-existing STIs and genital ulcer disease (syphilis, HSV, chancroid) substantially increase per-act transmission in BOTH directions by recruiting target cells and breaching mucosa. Ignoring concurrent STI status as a risk multiplier is the trap.



Biting/spitting — negligible

WHAT YOU SEE

The very bottom row of the league table:
BITING/SPITTING/TEARS — NEGLIGIBLE, NOT SIGNIFICANT — dead last because saliva does not transmit HIV.

WHAT IT ENCODES

Saliva, biting, and spitting are NOT significant routes of HIV transmission — saliva contains inhibitory factors and very low viral concentrations. These rank at the bottom of the league table as effectively negligible exposures in normal circumstances.

THE BOARD TRAP

WRONG: 'A human bite or being spat on requires HIV PEP.'
Saliva does not transmit HIV; PEP is not indicated for spitting or typical bites. The rare theoretical exception requires fresh blood in the saliva with a deep bite — but plain saliva exposure is the classic distractor that does not warrant prophylaxis.

RIGHT SIDE

TRANSMISSION RISK LEAGUE TABLE	
ROUTE	ROUTES
TOP SPOT	 BLOOD TRANSFUSION 9,250/10,000 = 92.5% — HIGHEST RISK 
	RECEPTIVE ANAL INTERCOURSE 138/10,000 — ~1.4% — THIN RECTAL MUCOSA — MICROTRAUMA — LANGERHANS CELLS DIRECTLY BELOW SURFACE
	NEEDLE SHARING



U=U flag

WHAT YOU SEE

A green banner shaped like a flag reading U=U, UNDETECTABLE = UNTRANSMITTABLE — the banner is the slogan that suppressed viral load means zero sexual transmission.

WHAT IT ENCODES

U=U (Undetectable = Untransmittable): a person on ART with a sustained undetectable plasma viral load does NOT sexually transmit HIV. This is a landmark, evidence-based prevention principle (PARTNER/PARTNER2 studies showed zero linked transmissions). It reinforces ART adherence and reduces stigma, and underpins treatment-as-prevention.

THE BOARD TRAP

WRONG: 'U=U means the patient can safely stop ART or skip condoms in all situations.' U=U applies only to SEXUAL transmission and only with SUSTAINED viral suppression on continuous ART — it is not a license to stop ART (viral load would rebound) and does not protect against other STIs. The discriminator: sustained suppression on ongoing therapy, sexual route only.

	RECEPTIVE ANAL INTERCOURSE 138/10,000 — ~1.4% — THIN RECTAL MUCOSA — MICROTRAUMA — LANGERHANS CELLS DIRECTLY BELOW SURFACE
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	INSERTIVE ANAL INTERCOURSE 11/10,000
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VIRAL LOAD
<20 COPIES/mL

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ACTS WITHOUT
TRANSMISSION

Viral load <20 copies/mL

WHAT YOU SEE

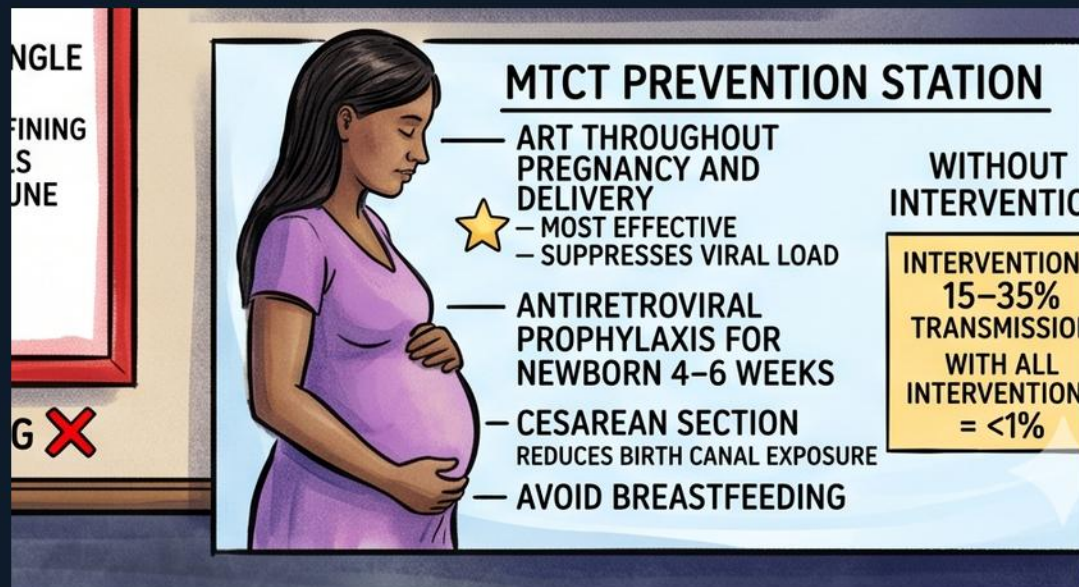
Green-banner text VIRAL LOAD <20 COPIES/mL = ESSENTIALLY ZERO SEXUAL TRANSMISSION, even with >125,000 sex acts — the tiny number drives the U=U promise.

WHAT IT ENCODES

Durable virologic suppression (commonly <200 copies/mL, ideally undetectable, e.g., <20–50 copies/mL by assay) is the threshold underlying U=U; at this level there is effectively zero sexual transmission. Most guidelines define 'undetectable for U=U' as sustained <200 copies/mL on ART, confirmed over time.

THE BOARD TRAP

WRONG: 'One undetectable viral load result makes the patient untransmittable.' A single undetectable result is insufficient — U=U requires SUSTAINED suppression (typically confirmed undetectable for ≥6 months and maintained on adherent ART). A lone result, or a recent ART start, does not yet confer U=U protection.



MTCT Prevention Station

WHAT YOU SEE

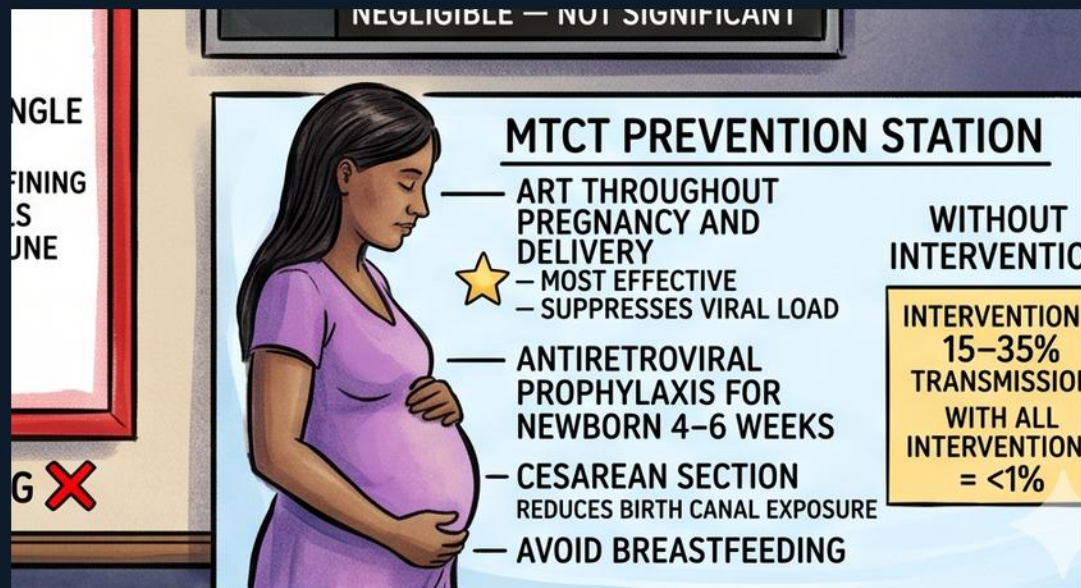
The MTCT PREVENTION STATION panel: a pregnant patient beside a list of interventions, contrasting WITH interventions <1% vs WITHOUT intervention 15–35% transmission — the pregnant figure anchors mother-to-child prevention.

WHAT IT ENCODES

Mother-to-child (perinatal/vertical) transmission is highly preventable through a bundle: maternal ART throughout pregnancy and delivery, appropriate delivery mode based on viral load, neonatal antiretroviral prophylaxis, and avoidance of breastfeeding where formula is safe. Maternal viral suppression at delivery is the single most important protective factor.

THE BOARD TRAP

WRONG: 'Without intervention, perinatal transmission is uncommon.' Untreated MTCT risk is high — about 15–35% (including breastfeeding) — but full intervention drops it to <1–2%. Underestimating the baseline untreated risk, or assuming one component (e.g., C-section alone) suffices, is the trap; it's the combined bundle that achieves <1–2%.



ART throughout pregnancy & delivery

WHAT YOU SEE

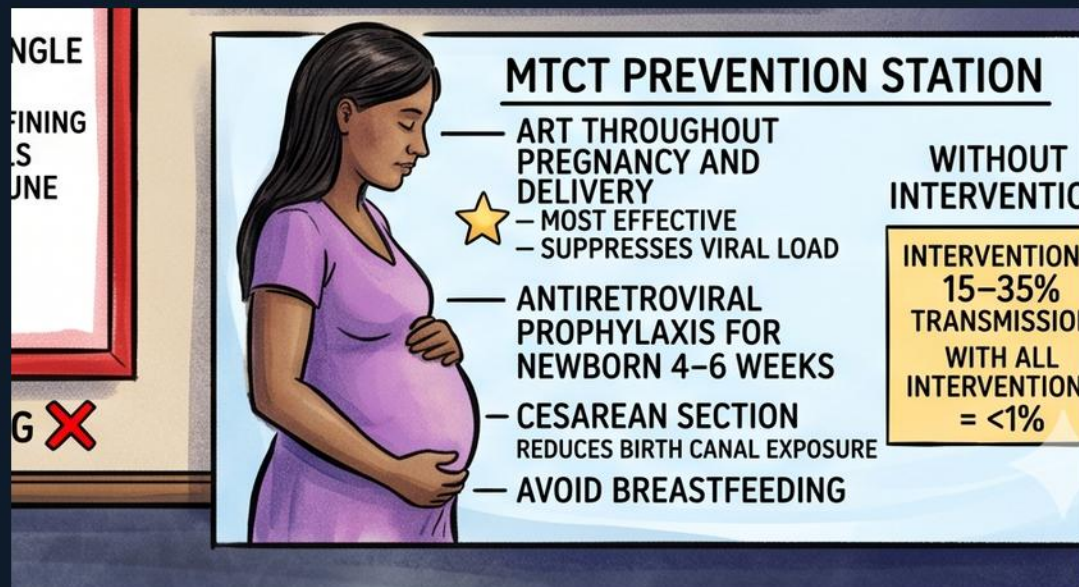
Top line of the MTCT station, gold-star bulleted: ART THROUGHOUT PREGNANCY AND DELIVERY — SUPPRESSES VIRAL LOAD — the gold star marks maternal ART as the single most protective step.

WHAT IT ENCODES

Start or continue ART in ALL pregnant women living with HIV regardless of CD4, with the goal of an undetectable viral load by delivery. Integrase inhibitor-based regimens (e.g., dolutegravir) are now preferred in pregnancy for rapid suppression. Suppressed maternal viral load is the dominant determinant of preventing vertical transmission.

THE BOARD TRAP

WRONG: 'Stop or pause ART during the first trimester due to teratogenicity concerns.' Do NOT delay or interrupt ART in pregnancy — interruption raises viral load and transmission risk. Modern preferred regimens (including dolutegravir) are considered safe; suppressing maternal VL outweighs theoretical drug concerns. Stopping ART is the trap.



Infant antiretroviral prophylaxis

WHAT YOU SEE

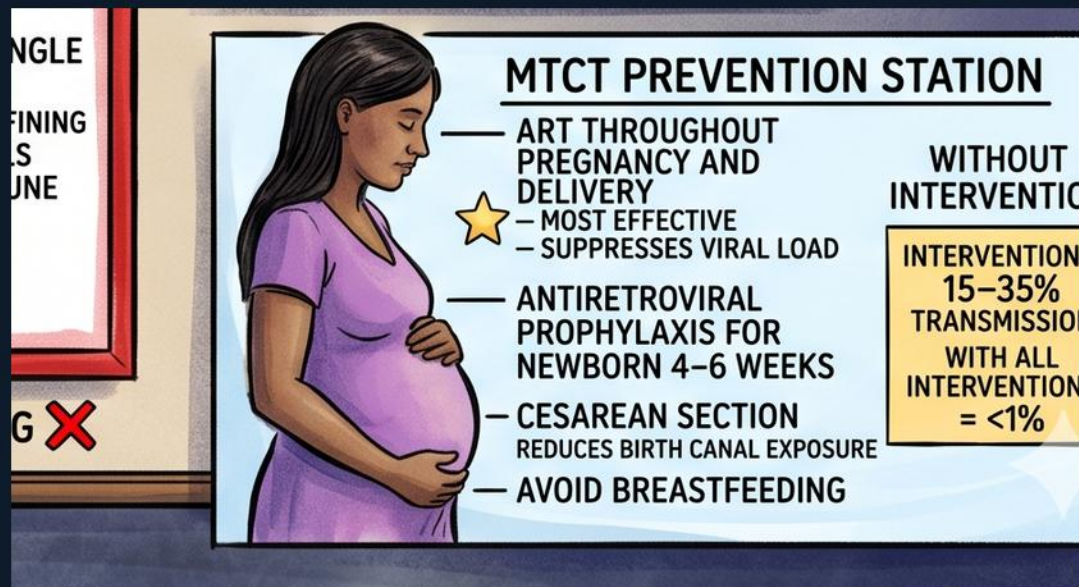
MTCT station bullet ANTIRETROVIRAL PROPHYLAXIS FOR NEWBORN 4-6 WEEKS — the newborn gets its own short ART course after birth.

WHAT IT ENCODES

Neonatal post-exposure prophylaxis is given for ~4–6 weeks after birth (zidovudine for low-risk; multi-drug regimens for higher-risk exposures) to further reduce transmission. The infant is monitored with HIV nucleic acid (DNA/RNA PCR) testing at intervals, because maternal antibodies cross the placenta and make infant antibody tests unreliable until ~18 months.

THE BOARD TRAP

WRONG: 'Diagnose HIV in an exposed newborn with an antibody test.' Infant antibody tests are unreliable due to transplacental maternal IgG until ~18 months — use HIV nucleic acid (PCR) testing for early infant diagnosis. Also, infant prophylaxis complements, and does not replace, maternal ART. Relying on infant serology is the trap.



Cesarean if VL elevated

WHAT YOU SEE

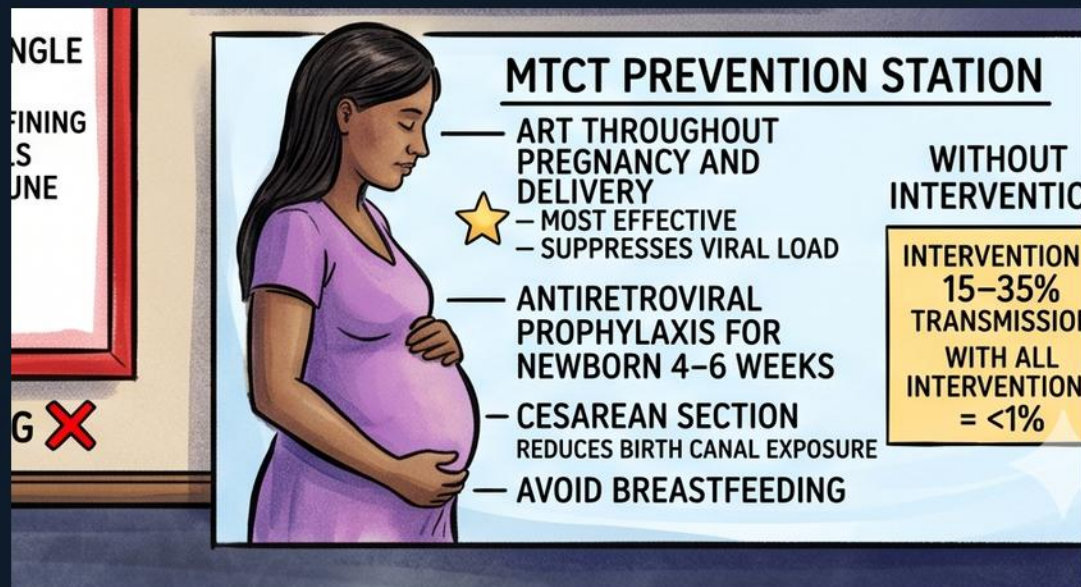
MTCT station bullet **CESAREAN SECTION** — REDUCES BIRTH CANAL EXPOSURE — C-section only when maternal viral load is not suppressed.

WHAT IT ENCODES

Scheduled (pre-labor, pre-rupture) cesarean section at ~38 weeks reduces intrapartum transmission when the maternal viral load is NOT suppressed (typically >1000 copies/mL near delivery). Add intrapartum IV zidovudine when VL is elevated/unknown. If VL is suppressed/undetectable, vaginal delivery is appropriate.

THE BOARD TRAP

WRONG: 'All HIV-positive mothers should deliver by cesarean.' C-section is only beneficial when maternal VL is >1000 (or unknown) near delivery; with suppressed/undetectable VL, scheduled C-section confers no added benefit and vaginal delivery is acceptable. Recommending universal C-section ignores the viral-load-dependent indication.



Avoid breastfeeding

WHAT YOU SEE

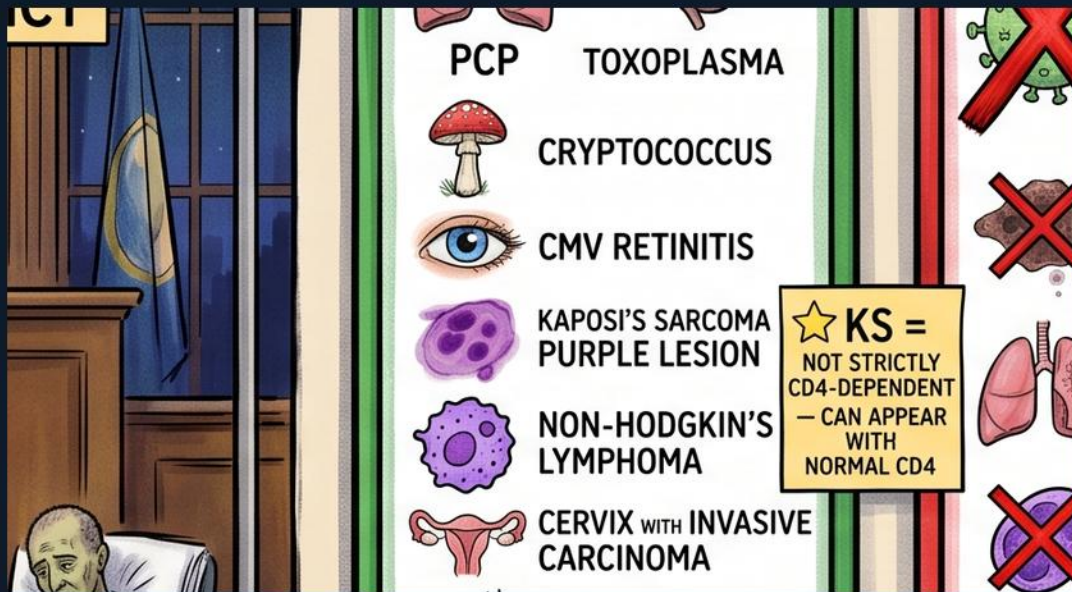
MTCT station bullet AVOID BREASTFEEDING — breast milk is an infectious route where safe formula is available.

WHAT IT ENCODES

In resource-rich settings, formula feeding is recommended because breast milk transmits HIV and safe alternatives exist. Breastfeeding accounts for a substantial share of vertical transmission when mothers nurse. The recommendation is context-dependent on the availability of safe, sustainable replacement feeding.

THE BOARD TRAP

WRONG: 'Breastfeeding is always contraindicated with maternal HIV everywhere.' In resource-LIMITED settings where safe water/formula is unavailable, WHO advises exclusive breastfeeding WITH maternal ART, because the risk of formula-related infant mortality outweighs residual transmission risk. The answer is setting-dependent, not absolute.



PrEP — tenofovir/emtricitabine

WHAT YOU SEE

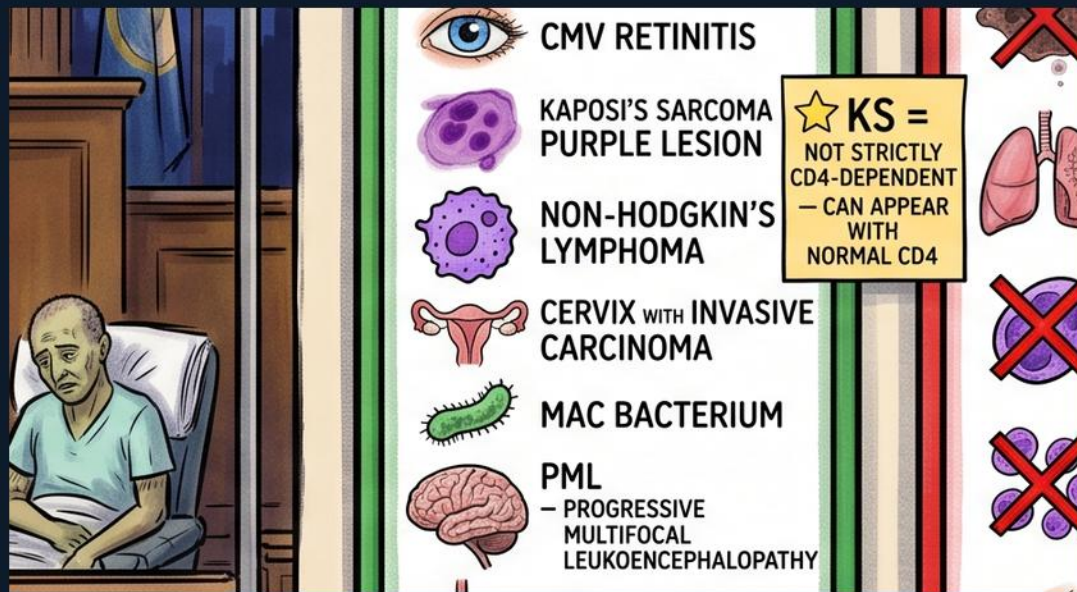
prevention panel — pre-exposure prophylaxis for HIV-negative high-risk individuals

WHAT IT ENCODES

Pre-exposure prophylaxis (PrEP): daily oral tenofovir/emtricitabine (TDF/FTC, or TAF/FTC) for HIV-NEGATIVE people at substantial risk (serodiscordant partners, condomless sex with high-risk partners, recent STI, PWID). Long-acting injectable cabotegravir every 8 weeks is an alternative. PrEP is highly effective (>90% with adherence). Confirm HIV-negative status before starting and re-test every ~3 months; monitor renal function.

THE BOARD TRAP

WRONG: 'Start PrEP without confirming the patient is HIV-negative.' You MUST exclude established HIV first (including acute HIV via 4th-gen Ag/Ab and, if acute infection suspected, RNA) — giving two-drug PrEP to someone with undiagnosed HIV risks resistance. PrEP is for HIV-negative people only; starting it in an undiagnosed positive is the trap.



PEP — within 72 hours

WHAT YOU SEE

prevention panel — post-exposure prophylaxis after high-risk exposure

WHAT IT ENCODES

Post-exposure prophylaxis (PEP): after a high-risk exposure (occupational needlestick, sexual assault, condom failure with a positive/unknown source), start a 3-drug ART regimen (e.g., tenofovir/emtricitabine + dolutegravir or raltegravir) ASAP and within 72 hours, continued for 28 days. Obtain baseline HIV testing and follow-up testing; assess source when possible.

THE BOARD TRAP

WRONG: 'Begin PEP whenever the patient presents, even days later.' PEP must be started within 72 hours of exposure — beyond that window efficacy drops sharply and it is generally not offered. Also, a patient with ongoing/repeated exposures should be transitioned to PrEP, not serial PEP courses. Starting PEP late, or using PEP for ongoing risk instead of PrEP, is the trap.

Summary — every symbol

1. Transmission Risk League Table

Per-exposure HIV transmission risk varies by orders of magnitude by route, so board questions often ask you to rank exposures

3. Receptive anal intercourse

Receptive anal intercourse is the highest-risk SEXUAL exposure (~138/10,000 acts, ~1.4%) because the rectal mucosa is thin, single-layered, richly vascular, and easily traumatized, giving the virus direct access to target cells

5. Receptive vaginal intercourse

Receptive vaginal intercourse carries higher risk (~8/10,000) than insertive vaginal intercourse (~4/10,000)

7. U=U flag

U=U (Undetectable = Untransmittable): a person on ART with a sustained undetectable plasma viral load does NOT sexually transmit HIV

9. MTCT Prevention Station

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11. Infant antiretroviral prophylaxis

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15. PEP — within 72 hours

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2. Blood transfusion — top risk

Transfusion of HIV-infected blood is the single highest per-exposure transmission risk, roughly 90%+ (about 9,250+/10,000 acts)

4. Needlestick — occupational

Occupational percutaneous needlestick from an HIV-positive source carries ~0.3% risk (~23/10,000)

6. Biting/spitting — negligible

Saliva, biting, and spitting are NOT significant routes of HIV transmission — saliva contains inhibitory factors and very low viral concentrations

8. Viral load <20 copies/mL

Durable virologic suppression (commonly <200 copies/mL, ideally undetectable, e.g., <20–50 copies/mL by assay) is the threshold underlying U=U; at this level there is effectively zero sexual transmission

10. ART throughout pregnancy & delivery

Start or continue ART in ALL pregnant women living with HIV regardless of CD4, with the goal of an undetectable viral load by delivery

12. Cesarean if VL elevated

Scheduled (pre-labor, pre-rupture) cesarean section at ~38 weeks reduces intrapartum transmission when the maternal viral load is NOT suppressed (typically >1000 copies/mL near delivery)

14. PrEP — tenofovir/emtricitabine

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